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Degree	University	Stream	Year	CPI/%
MS by Research	IIT Bombay	Computer Science (AI/ML)	2023 - 2026	8.83
B.Tech	Amity University	Computer Science & Engineering	2015 - 2019	7.65

PUBLICATIONS

- **Aakash Kumar Agarwal**, Om Chaudhari, Nihar Ranjan Sahoo, Abhinav Joshi, Ashutosh Modi. *Learning to Forget: Hypernetwork-Driven Sharded Unlearning in LLMs*, under review at **ACL ARR 2026**.
- **Aakash Kumar Agarwal***, Moyuru Yamada*. *Hypothesis-Driven Reasoning for Large Language Models*, Accepted at **AAAI 2026** (Main Conference). (*Equal contribution)
- **Aakash Kumar Agarwal***, Saprativa Bhattacharjee*, Pushpak Bhattacharyya. *ReDepress: A Cognitive Framework for Detecting Depression Relapse from Social Media*, Accepted at **EMNLP MAIN 2025**. (*Equal contribution)

WORK EXPERIENCE

- **AI Research Engineer Intern | Fujitsu Research, Bangalore** (May 25 - July 25)
Guide: *Moyuru Yamada, Fujitsu Research*
 - Developed a **Hypothesis-Driven Reasoning (HDR)** framework to improve LLM reasoning via iterative hypothesis generation and verification over multimodal data.
 - Built a two-stage memory consolidator with **LangChain** and **LangGraph**, combining automatic factor extraction with iterative refinement via a generate-and-verify loop using MLLMs.
 - Introduced an *Experience-Transfer* task to test compositional generalization across unseen multi-modal contexts. Evaluated HDR on multiple multimodal LLMs (GPT-4o, Qwen2.5, Gemma3), achieving near-oracle performance, **improving accuracy by up to 56.7% over baselines**.
- **Student AI Researcher | CFILT Lab | IIT Bombay, Mumbai** (Jan 24 - Present)
Guide: *Prof. Pushpak Bhattacharyya, IIT Bombay*
 - Researching applications of NLP in mental health, using social media data.
 - Investigating LLM safety under cognitive load and multilingual adversarial conditions.
- **Software Developer | Accenture, Gurugram** (May 19 - July 21)
 - Developed and maintained software applications to meet client requirements and business needs.
 - Collaborated with cross-functional teams to implement new features and enhance existing functionalities.
 - Participated in the full software development lifecycle, from requirements gathering to deployment.

RESEARCH PROJECTS

- **Machine Unlearning for LLMs via HyperLoRA Adapters** (Jan 25 - Present)
R&D Project | Guide: *Prof. Ashutosh Modi (Exploration Lab, IIT Kanpur)*
 - Developed **HyperLoRA**, a hypernetwork that generates per-shard LoRA adapters from shard embeddings, enabling **instant, auditable LLM unlearning** via constant-time ($\mathcal{O}(1)$) embedding deletion—base model frozen, no gradient step.
 - Designed and evaluated **5 adapter composition/routing strategies** (*merged, z_emb_avg, per_z, router_lookup, router_probe*) on TOFU with **LLaMA-3.2-1B-Instruct**; the deployable router hit **0.80 AGG**, **matching the retrain-from-scratch gold (0.81 AGG)** and far above the strongest tuned post-hoc baseline (**0.53 AGG**); robust across shardings (**K=20→40**) and scale (**1B→3B**).
- **Detecting Depression Relapse Using Cognitive Biases** (June 24 - Dec 25)
MS Thesis | Guide: *Prof. Pushpak Bhattacharyya (CFILT Lab, IIT Bombay)*
 - Predicted depression relapse from post-remission social media behavior using cognitive bias features and temporal modeling.
 - Constructed dataset from 10 years of Reddit dumps using regex filtering and multi-stage LLM-based filtering (LLaMA 3.2 3B, LLaMA 3.1 70B), followed by clinical expert annotation.
 - Engineered cognitive bias features and performed statistical analysis confirming relapse-bias correlation.
 - Implemented temporal modeling with transformer model on chronologically ordered post embeddings, improving **accuracy from 0.81 to 0.88** and **F1 from 0.76 to 0.84** by integrating cognitive biases with post embeddings.
- **Cognitive Load and Multilingual Vulnerabilities in LLM Jailbreaking** (Jan 25 - Dec 25)
R&D Project | Guide: *Prof. Pushpak Bhattacharyya (CFILT Lab, IIT Bombay)*
 - Analyzed how cognitive load and adversarial context affect LLM safety in English and four Indian languages.
 - Demonstrated that adding **cognitive load reduces LLMs' attention to safety signals**, and developed four strategies that effectively increase cognitive load and bypass safety mechanisms.

COURSE PROJECTS

• Personalization of LLM Based on Profession

Human Centered AI: From Learning Models to Responsible Systems, Spring 2025

- Developed a profession-aware LLM personalization framework, conditioning responses on both profession and context using **132 profession-subprofession pairs** and a dataset of **150K profession-question preference pairs** for contextual alignment and ranking.
- Fine-tuned Gemma-2-2B with LoRA, used DPO/GRPO alignment to optimize profession-conditioned responses, improving personalization quality.

• GradSafe: Gradient-Based Jailbreak Detection in LLMs

Speech and Natural Language Processing and the Web, Autumn 2024

- Developed **GradSafe**, a gradient-based jailbreak detection method achieving **>70% F1 score** on adversarial datasets by analyzing safety-critical parameters, and enhanced performance with Logistic Regression and SVM classifiers using **PCA (87K → 1K features)**, yielding **5-8% accuracy improvement**.
- Curated and evaluated a dataset of 100 persuasive adversarial jailbreak prompts, testing LLaMA-2 variants and exposing alignment limitations using GradSafe.

• Image Retrieval Using Neural Network based Hashing

Information Retrieval & Mining for Hypertext & the Web, Autumn 2024

- Explored CIFAR-10 and CIFAR-100 datasets for the image retrieval task using **Locality-Sensitive Hashing (LSH)** based on Neural Networks, with image embeddings extracted from a Vision Transformer (ViT).
- Achieved strong results with one-stage training of the hash code: **CIFAR-10 (mAP: 90%, Precision@10: 89.7%)** and **CIFAR-100 (mAP: 76%, Precision@10: 84%)** compared to Random LSH baselines.
- Further improved performance with two-stage training of embedding and then hash code: **CIFAR-10 (mAP: 96%, Precision@10: 95.5%)** and **CIFAR-100 (mAP: 87%, Precision@10: 84%)**.

• Explainable Malware Classification

Foundations of Machine Learning, Autumn 2023

- Built **ResNet-50** and **EfficientNetV2** for multiclass malware detection by converting binaries to grayscale images, using Microsoft Malware Classification Challenge and Mali-MG datasets, achieving **97.25% / 94.85%** (Microsoft) and **98.71% / 97.58%** (Mali-MG), with ResNet-50 matching **state-of-the-art performance**.
- Applied **SHAP values** to the models for interpretability of **feature-level insights** into malware classification.

TECHNICAL SKILLS

- **AI & ML:** Transformers, LLMs, Fine-tuning, LoRA, Prompt Engineering, RAG (Vector Database), Safety Alignment, Hugging Face, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, LangChain, AI Agents (MCP, Tool Use)
- **Programming Languages:** Python, C++
- **Web & Database:** SQL, HTML, CSS
- **Tools & DevOps:** Shell, Git, GitHub, Docker, FastAPI

KEY COURSES TAKEN

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|--|-----------------------------------|
| • Speech and Natural Language Processing and the Web | • Foundations of Machine Learning |
| • Deep Learning for Natural Language Processing | • Automatic Speech Recognition |
| • Information Retrieval & Mining for Hypertext & the Web | • Algorithms and Complexity |
| • Human-Centered AI | • Software Lab |

POSITIONS OF RESPONSIBILITY

- **Co-Lead | Inter-IIT Tech Meet 14.0** (Oct 25 - Dec 25)
 - Co-led a 20-member team to develop a real-time Dynamic Trust Profile system integrating KYC, credit risk, and fraud analytics.
 - Designed core architecture across streaming pipelines, incremental ML, and LLM-driven decisioning for financial intelligence.
- **Teaching Assistant | Organisation of Web Processing** (Prof. Soumen Chakrabarti) (Aug 24 - Dec 24)
 - Assisted in invigilation, grading, and exam management to ensure smooth course execution.
- **Teaching Assistant | Computer Programming and Utilization** (Prof. Shivaram) (Jan 24 - May 24)
 - Collaborated with a team of 50+ TAs to support a class of 700+ students.
 - Supervised lab sessions, conducted help sessions, and fostered a productive learning environment.

SCHOLASTIC ACHIEVEMENTS

- Secured **All India Rank 156** in **GATE Computer Science 2023** among **75,680 candidates**

INTERESTS AND HOBBIES

- Programming, Badminton, Table Tennis, Chess